AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 23 · STUDENT LESSON

Block to Python Bridge

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Map the actions. Predict the state. Check the actual result.

Translate bounded paper procedures into Python while preserving order and evidence.

Keep one habit: Packing warm-up → five-action book loop → strict/empty boundaries → missing-field guard → independent sum/reset.

MAIN LESSON · SECTION 1

Can the same procedure survive a change of representation?

> Predict a short script, then keep observed output separate from the prediction.

Your learning goals

I can translate a short paper calculation into a Python script.

I can predict and explain the order of executed statements.

I can keep expected and observed **output** separate when checking a **program**.

Recall the calculation and interface

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Four packs contain six counters each. What steps find and show the total?

Your end goal

By the end, I can explain block to python bridge using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 23: Block-to-Python Bridge

Variables, **condition**s, **list**s, loops, and **function**s carry the same logic across representations.



CREATOR MISSION

Translate a block workflow into readable Python.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Translate the rule before adding syntax

> Use a teacher-prepared editor or label a paper trace clearly.

Connect blocks, state and code

A block workflow and Python can express the same ordered procedure. First identify the input values, state changes, conditions and displayed result. Assignment changes a stored value; print displays one. Indentation marks the statements controlled by a loop or condition. The packing program is a short warm-up; the later book-counting trace adds a loop and a strict boundary. Neither procedure trains an AI model.

Read the packing warm-up contract

In Grade 6 you traced rules on paper. Now you will write a small Python **program** and check what it actually does. Begin with a counter-packing calculation, not the whole activity adviser.

The requirement is to find the total counters in equal packs and display that total. For this first **program**, the supplied quantities are whole, nonnegative numbers; it does not yet validate typed **input**.

Run a saved script or trace on paper

1. Use a teacher-prepared Python 3 editor that can run a saved .py file. Open a new file called counters.py, type the code exactly as shown in the next section, save it, and use the editor’s Run command.

   Read the **output** area separately from the source editor. No account, AI tool, package installation or personal data is needed. If a device is unavailable, trace on paper and label the result predicted rather than executed.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use block-to-python bridge to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Trace four ordered packing statements

> The input contract supplies nonnegative whole counts; this warm-up does not validate arbitrary input.

Map the paper actions to code

A calculation from paper to output

PAPER STEP	PYTHON INSTRUCTION	EFFECT
Record pack count	packs = 4	Name holds 4
Multiply two quantities	total = packs * per_pack	Name holds product
Show the answer	print(total)	Displays the value

Assign per_pack before the multiplication. This diagram summarises the steps; the complete script below includes that assignment.

Complete packing program

```
packs = 4
per_pack = 6
total = packs * per_pack
print(total)
```

Explain assignment and displayed output

Read the script from top to bottom. The equals sign assigns the value on its right to the name on its left. The star means multiplication. print(total) displays the value of total; it does not display the letters in the name.

Type ordinary straight characters and keep the parentheses. The Python prompt symbols >>> are not part of this saved script.

Explain the deciding step

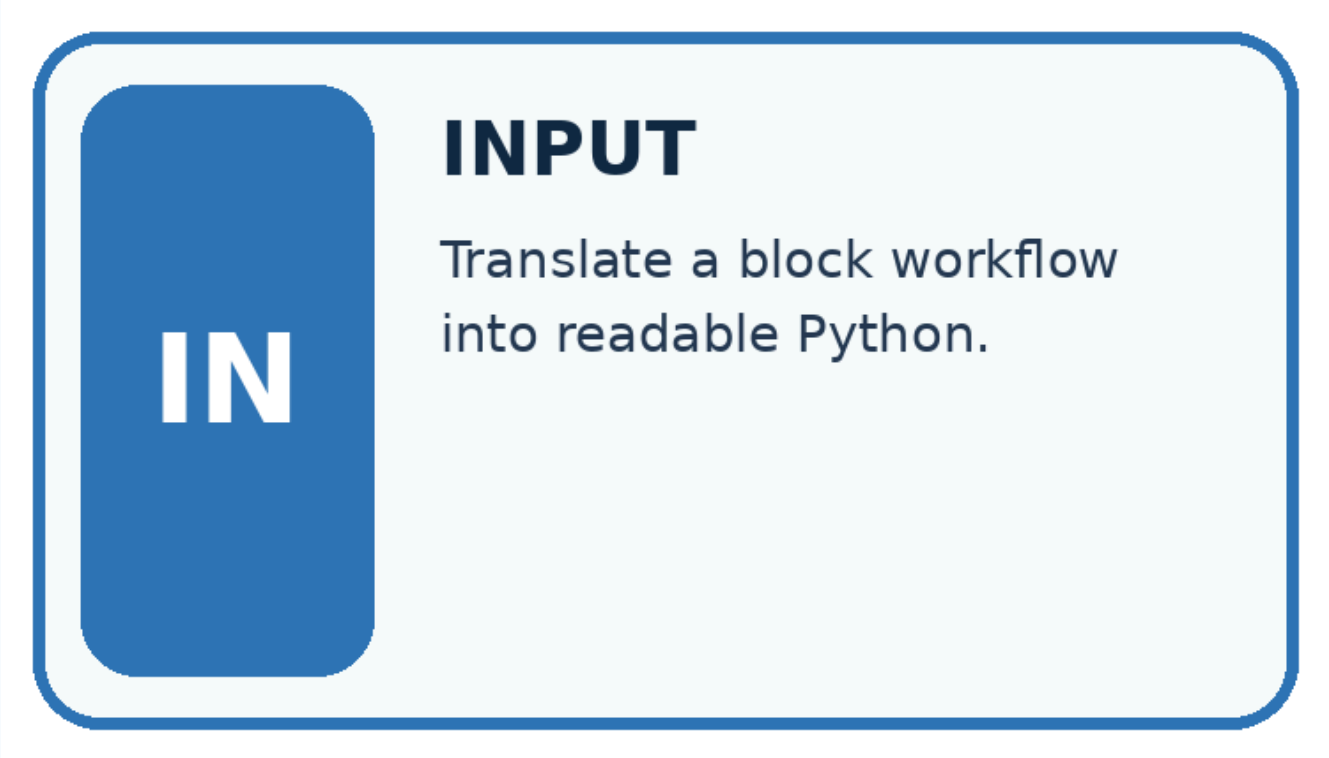
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Variables, conditions, lists, loops, and functions carry the same logic across representations.
↓
OUT
EVIDENCE
Translate a block workflow into readable Python.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

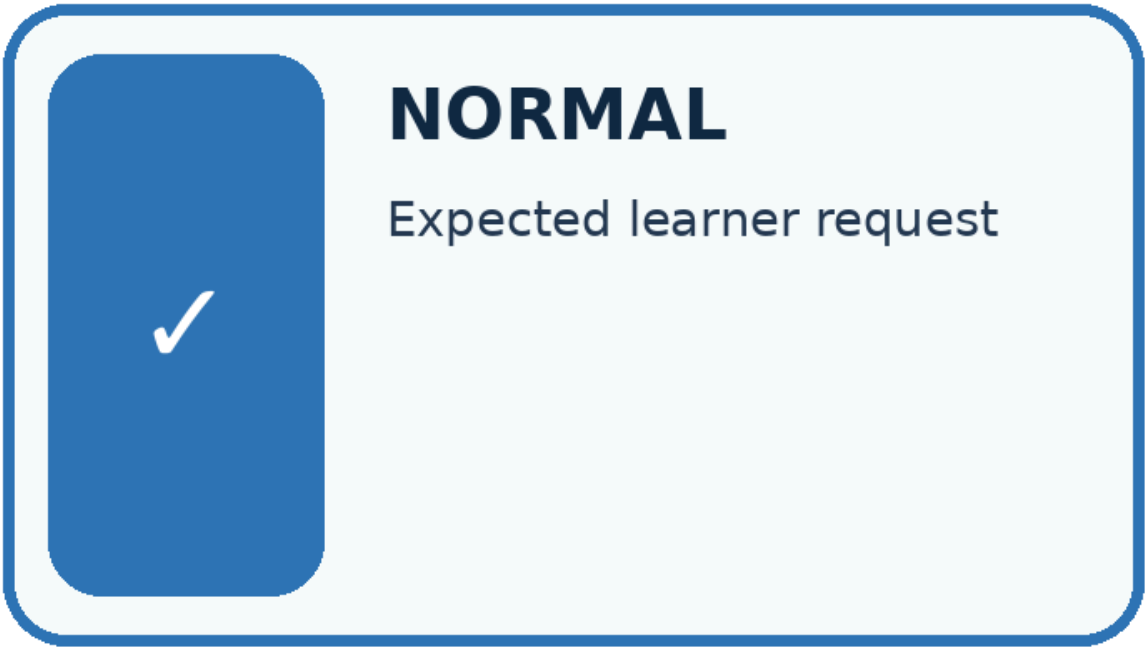
Connect the picture and the explanation

MAIN LESSON · SECTION 4

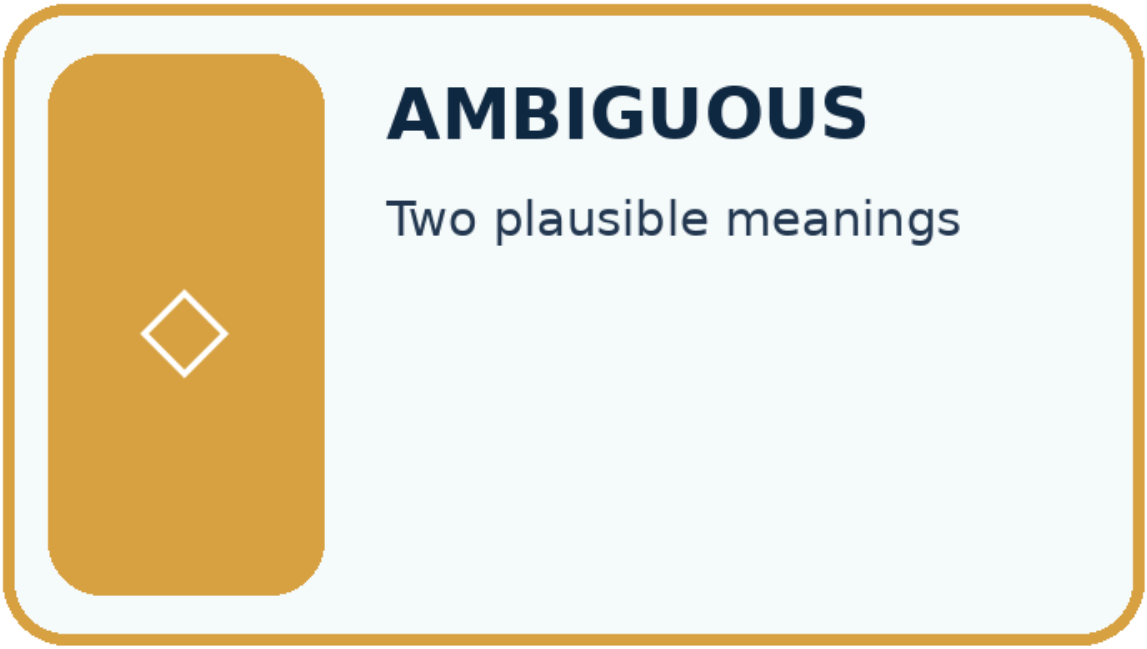
Compare a correct procedure with a misplaced statement

> Initialization, loop body and final output occupy different positions.

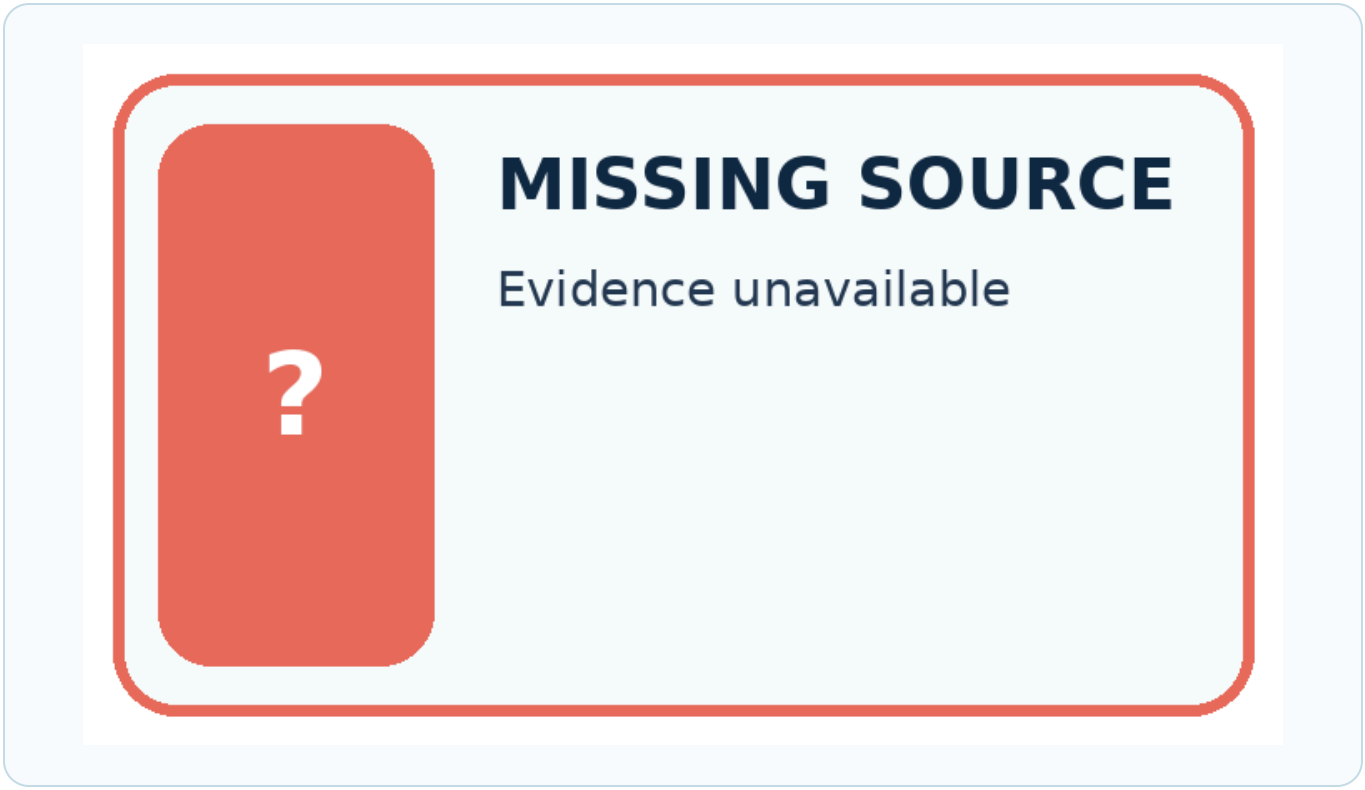
Four pictures with a purpose



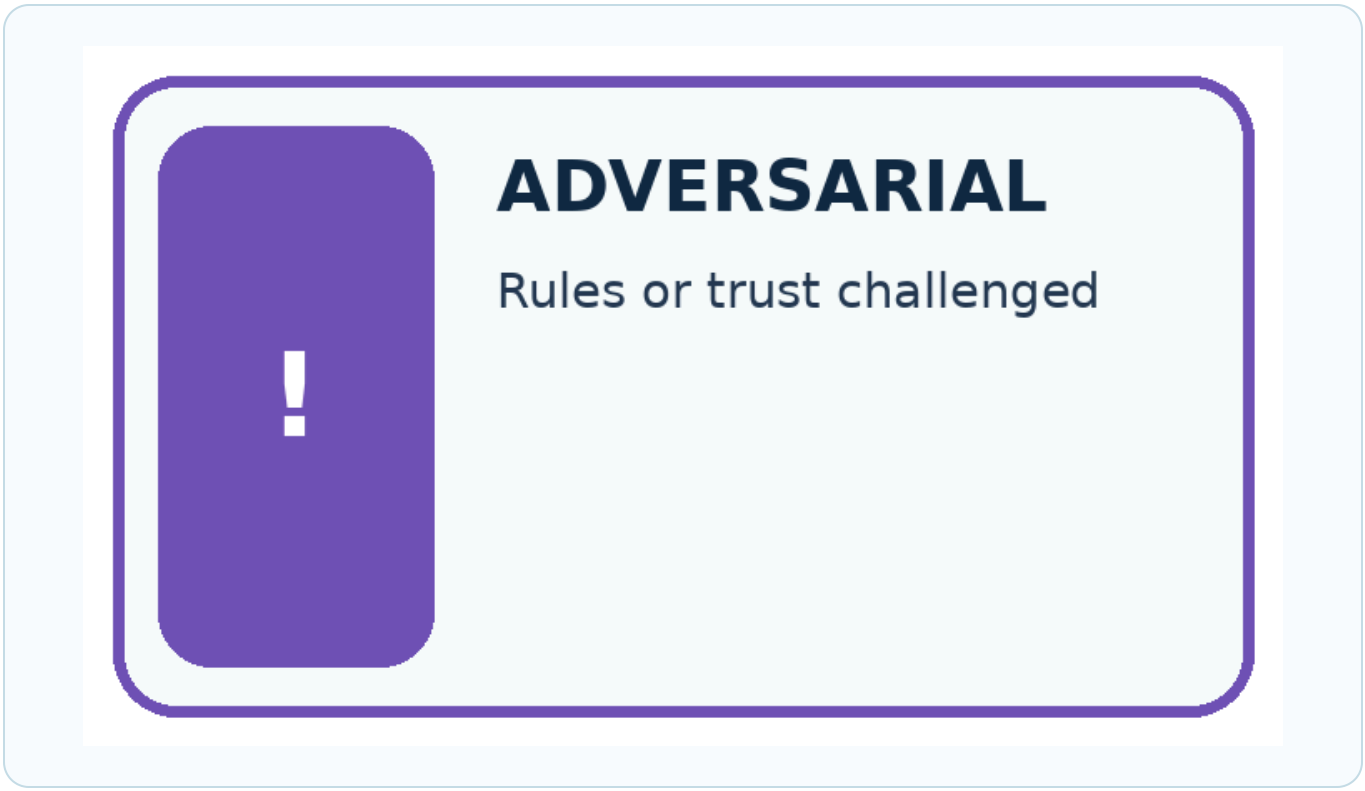
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Replace the last page count 90 with 100.

▼ Check after predicting

Prediction: The final count becomes 1.

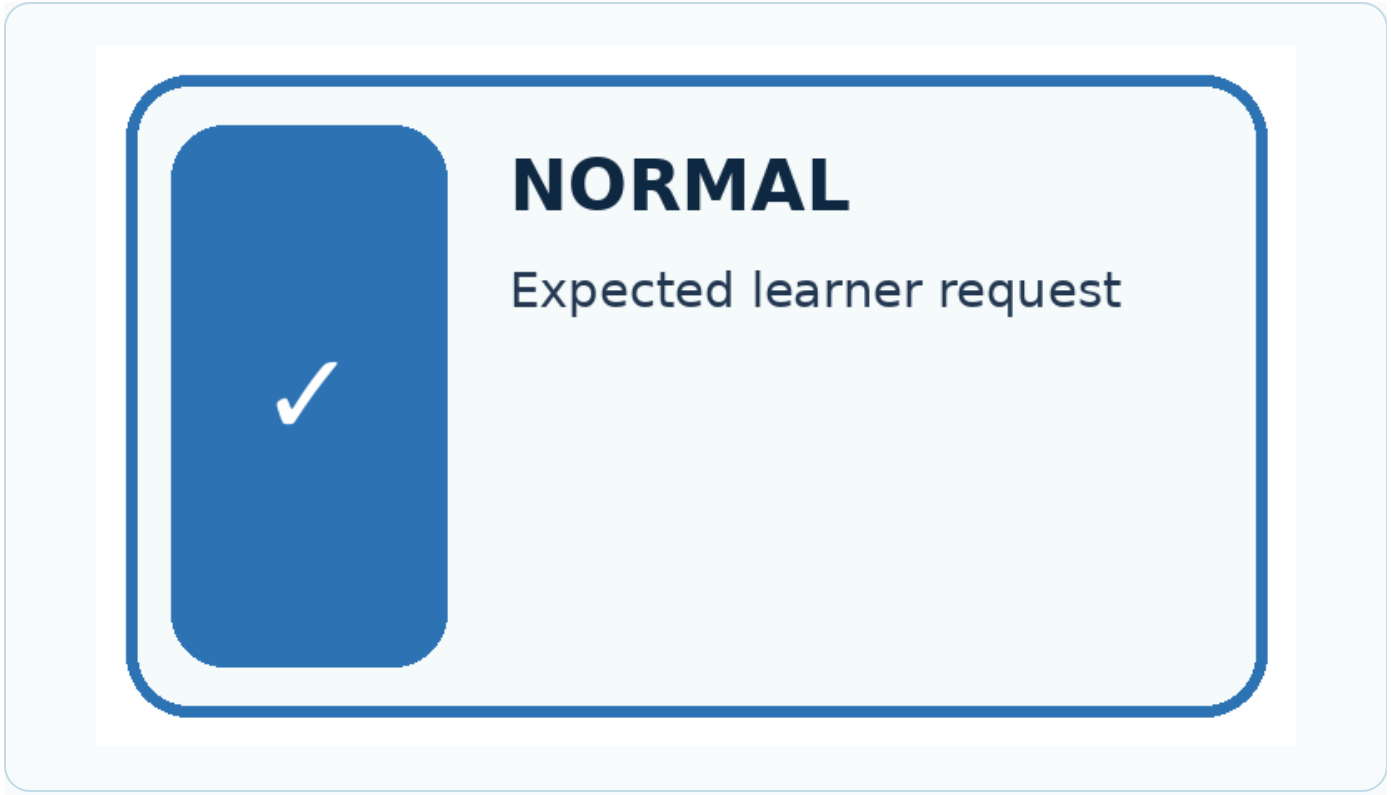
Reason: The comparison is strictly less than 100, so the boundary value does not qualify.

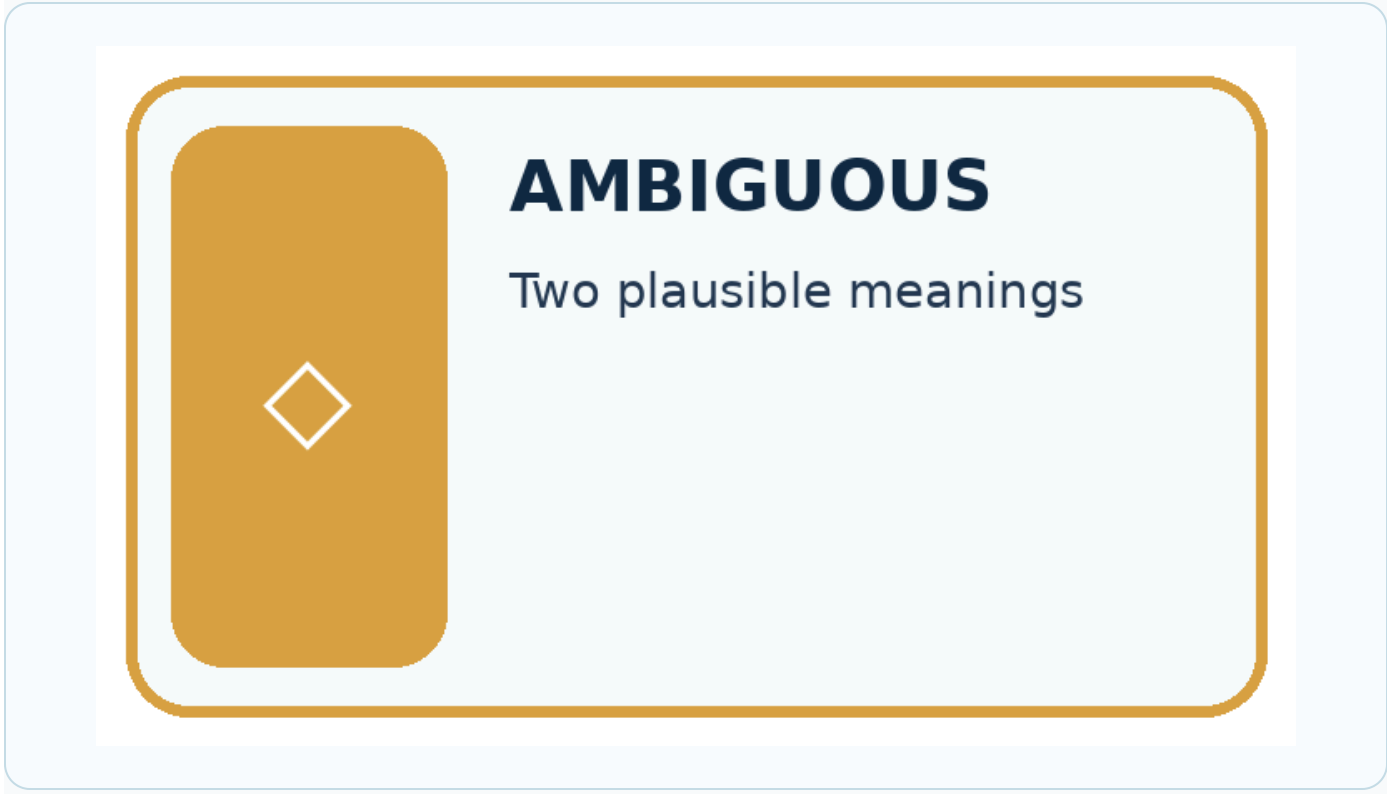
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the book counter one decision at a time

> Counting qualifying books differs from summing page counts.

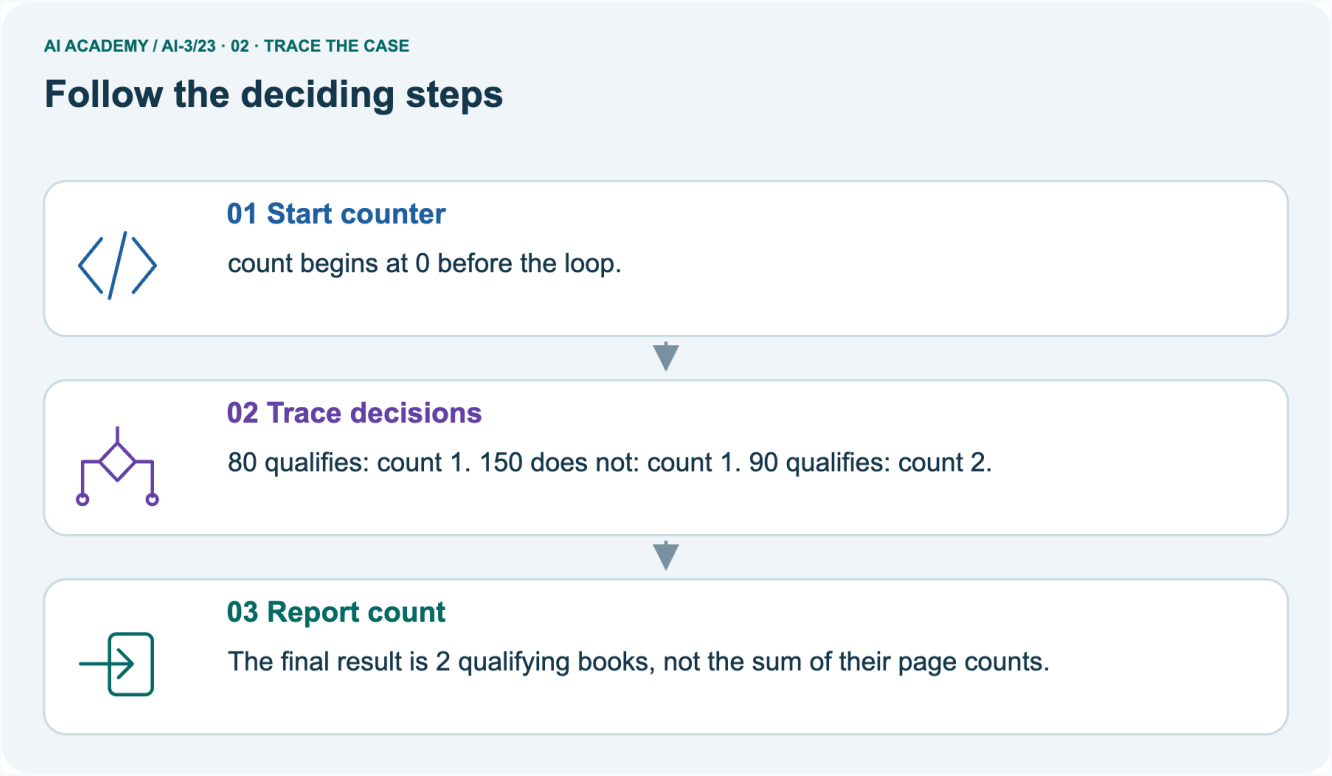
Read the complete book-count input

Translate a “count qualifying books” paper procedure into a Python-style trace.

Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Trace the labelled decisions



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Start counter	count begins at 0 before the loop.
2. Trace decisions	80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2.
3. Report count	The final result is 2 qualifying books, not the sum of their page counts.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Connect five block actions to executable code

Block action	Python and effect
Set counter once	count = 0 before the loop
Repeat for each book	for pages in page_counts:
Test the strict rule	if pages < 100:
Increase qualifying count	count = count + 1, inside the condition
Finish the function	return count after the loop; print the returned result

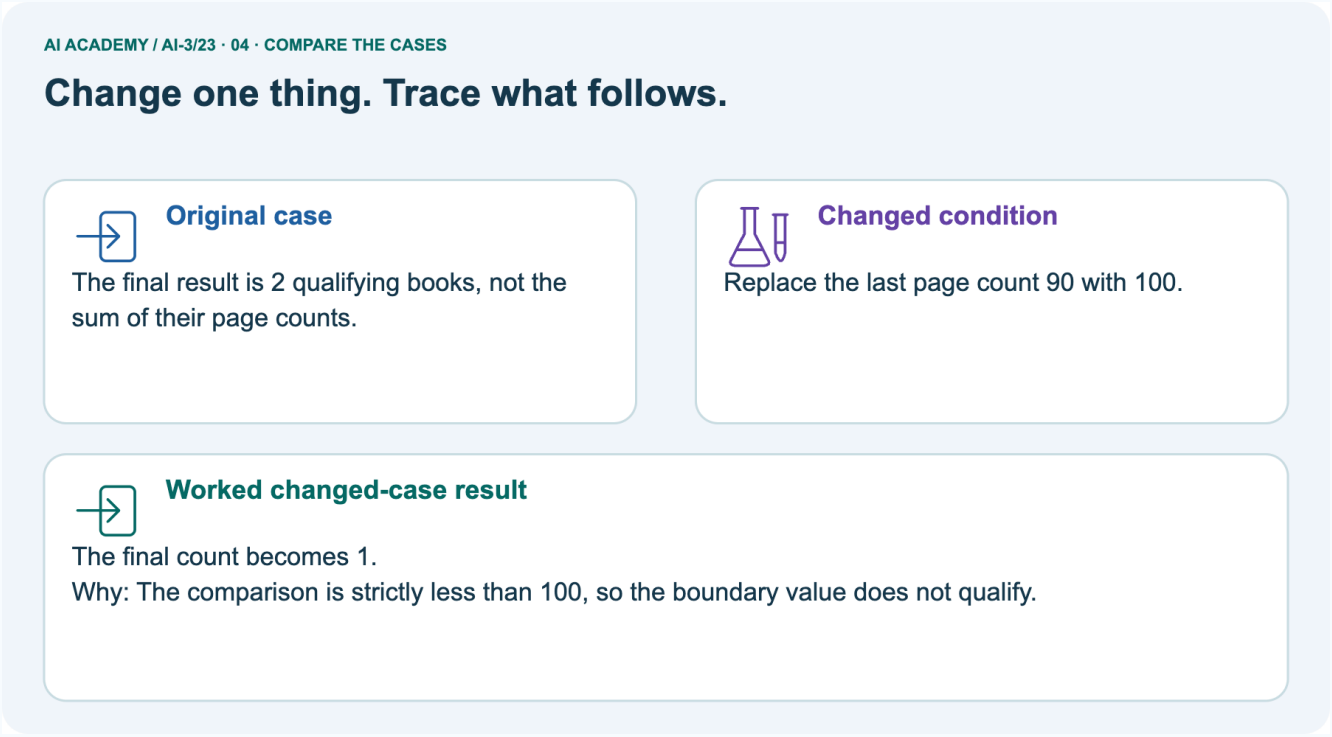
```
def count_short_books(page_counts):
    # Supplied input: a list of nonnegative integer page counts.
    count = 0
    for pages in page_counts:
        if pages < 100:
            count = count + 1
    return count

print(count_short_books([80, 150, 90]))
print(count_short_books([80, 150, 100]))
print(count_short_books([]))
```

Predict replacing 90 with 100

- Change one thing: Replace the last page count 90 with 100. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State what this deterministic result does not measure

- This deterministic loop counts a stated **condition**; it neither trains a model nor measures whether the books suit a reader.

- Read the labelled picture: Trace the changed **list** and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Packing warm-up · Prediction versus observation

Write 24 as the expected **output** before running. Then copy the actual **output** into a separate column. A matching result supports this case, not every possible **input**.

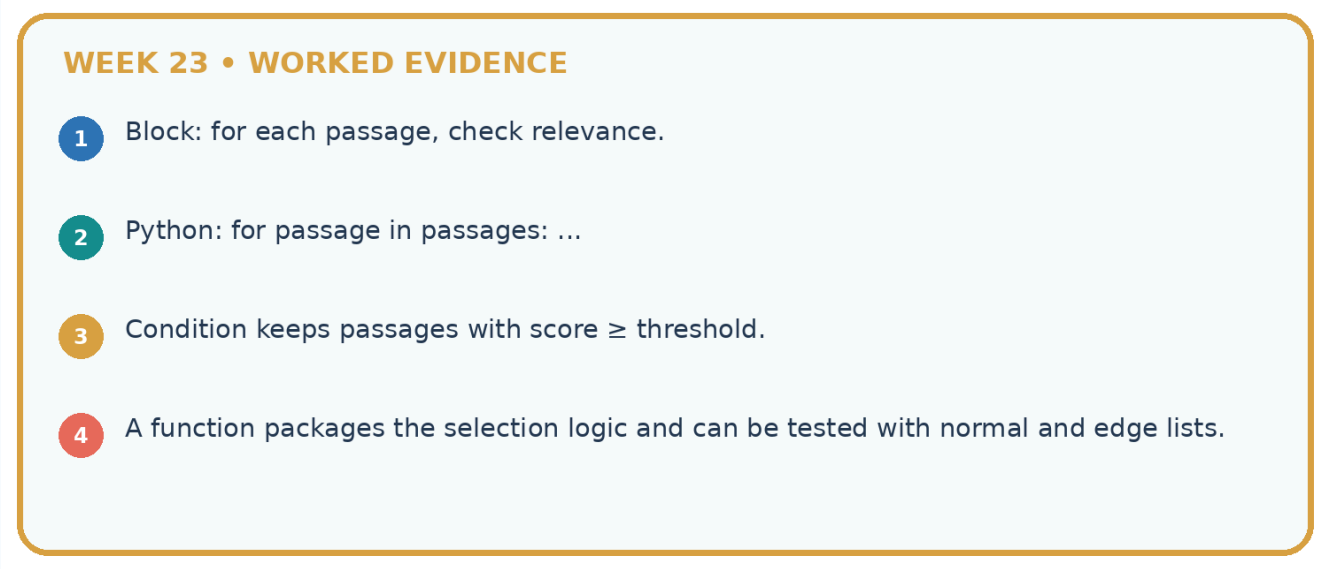
A saved file may contain an old edit: save again and rerun the whole script. Do not type the expected number into the **output** record without running.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Block: for each passage, check relevance.

2. Python: for passage in passages: ...

3. **Condition** keeps passages with score ≥ threshold.

4. A **function** packages the selection logic and can be tested with normal and edge **list**s.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

After line	Names and values	Displayed output
packs = 4	packs is 4	Nothing yet
per_pack = 6	packs is 4; per_pack is 6	Nothing yet
total = packs * per_pack	total is 24	Nothing yet
print(total)	Values are unchanged	24

> **Why is there no displayed output after the third line? What changes if print(total) is placed first in a fresh run, before any assignment? Explain using which names exist.**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Block to Python Bridge

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a statement placed at the wrong scope

> A passing example can hide a changed procedure.

Repair the reset and boundary shortcuts

Claim to repair: Put count=0 inside the loop so it starts fresh for every book. That erases the accumulated count before each decision. In the supplied three-book list it can accidentally finish at 1 instead of 2. Keep initialization before the loop. A different error is using pages<=100 when the rule says fewer than 100; boundary 100 must not qualify.

Explain the state change instead of fixing code until it happens to print the expected number.

Design a revealing follow-up

> **A script runs without an error and displays 11 for five packs of six. Is it correct for the stated requirement? Give the required result, the likely wrong operation and one check after repair. State whether your evidence came from a device run or a paper trace.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Moving blocks into text automatically creates correct Python. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Save the file with the **input** values, expected and observed **output**s, and a short explanation. This **program** follows a written multiplication rule.

It has not trained a model, predicted somebody's preferences or checked that real packs contain the stated number. Its conclusion depends on the supplied data. Week 2 examines the values and types behind these names.

MAIN LESSON · SECTION 8

Practise packing predictions with less help

> Use this simple script before the separate book loop.

Complete W2 prediction and observation

Use workbook W2 for the connected example “From a paper **algorithm** to Python”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Check assignment versus display

- Use the worked case for From a paper **algorithm** to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured changed list

- Trace the changed **list** and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Keep independent work separate

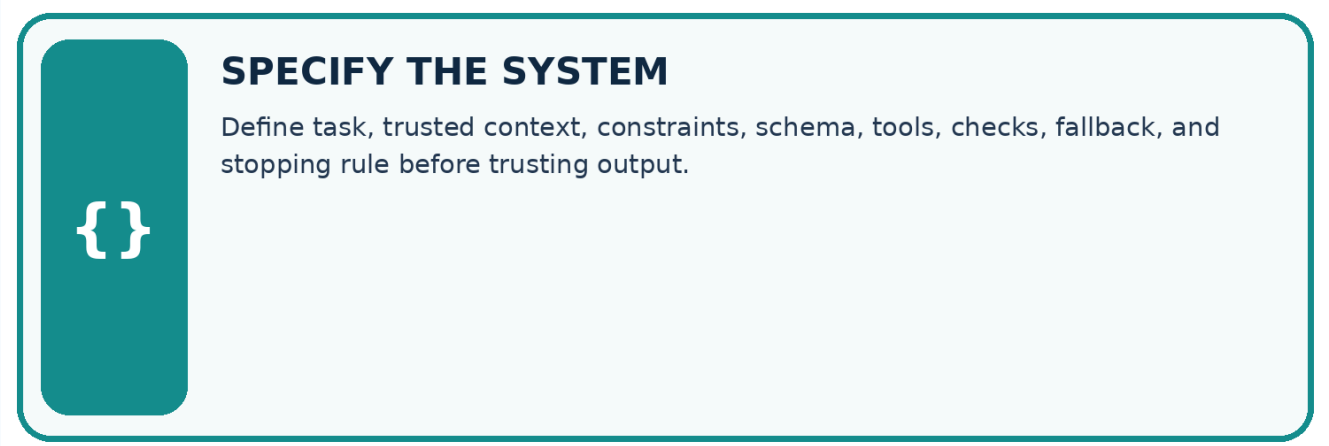
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Change only packs to 5. Predict the result from five groups of six, save, then run the entire file again. The expected **output** is 30. Keep the original four-pack result beside the new one and label the changed **input**.

A comment can explain an instruction: Python ignores text after # outside a string. A comment does not change the calculation.

```
packs = 5  # five equal packs
per_pack = 6
total = packs * per_pack
print(total)
```

A **program** can run successfully and still solve the wrong problem. Replacing * with + would display 11 for this **input**, but addition does not count five groups of six.

Use the requirement and a paper calculation to check meaning; the absence of an error message is not proof of correctness.

Practise with less help: Predict then observe

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Explain the picture

Test five complete block-to-code cases

> Separate the runnable numeric-list program from a proposed missing-field guard.

All supplied inputs and expected boundaries

Complete supplied case	Expected procedure/result
A · Page list[80,150,90]	Book-count trace0→1→1→2
B · Page list[80,150,100]	Strict boundary excludes100; result 1
C · Empty page list[]	No iterations; initialized count 0 returned
D · One book record {"title":"Cedar","pages":80}	pages field supplied; one qualifying book
E · One book record {"title":"Maple"}, pages absent	No page measurement; ask for it or report incomplete, not pages 0

Translate, predict, run or trace, then explain

1. Translate the five displayed block actions into the supplied Python function. Predict A/B/C outputs, then run or trace and label your evidence. D/E add a record-reading design question: D has a supplied page value; E does not. The numeric-list function is not a record parser, so do not pass those dictionaries into it.

   State a guard that checks the pages field before using it; missing data must not silently become zero. Record any actual errors rather than replacing them with predictions. Keep the independent sum/reset task withheld.

Keep the evidence record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

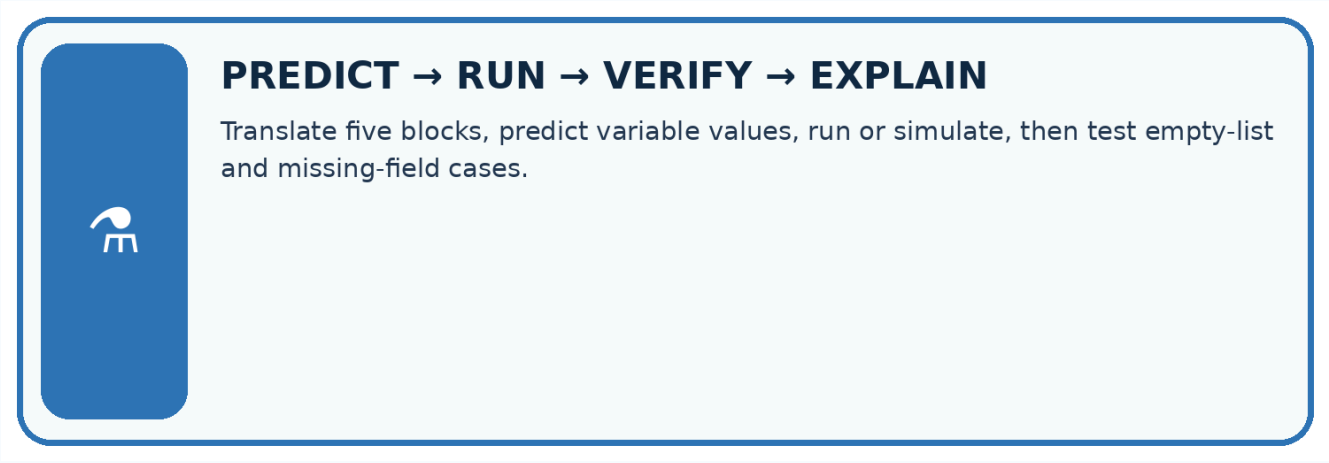
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the complete five-action book-count mapping and supplied cases A–E. Run or trace numeric **list**s, including empty and boundary cases. For record cards, explain the missing-pages guard separately; do not claim the numeric-**list** function parses records or validates arbitrary input.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: explain the fresh sum-and-reset procedure

> Use the independent values without substituting a practised packing case.

Independent procedure and scope

Set total=0, then for values2,4,9 perform total=total+value. Trace the result. What changes if total=0 moves inside the loop before that addition?

Trace both versions of the supplied sum procedure in order, keeping each intermediate total. Compare initialization before the loop with initialization inside it. This fresh task sums values; do not copy the book counter's increment-by-one rule.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Changed independent case

Week 23 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Trace this Python: score = 2; then if score >= 2: score = score + 3. What is the final value, and what changes if the **condition** becomes score > 2?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Implement small transparent **function**s for selection, validation, and citation formatting.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Write Python that totals [3,5,7], then trace each total. Explain how indenting 'total=0' inside the loop changes the result.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's code and evidence together

> Translate one stage with a clear input contract.

This week's contribution

For the Verified Study Buddy, translate one bounded deterministic stage into readable code before attempting a full assistant. Keep source values, assignment, decisions, output and input assumptions visible. Save the code version and distinguish predicted output, executed output, paper trace and error. A working loop is evidence for the supplied computation, not model training or learner benefit.

Retain the checkpoint and human choice

THIS WEEK'S CHECKPOINT Implement small transparent **function**s for selection, validation, and citation formatting.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

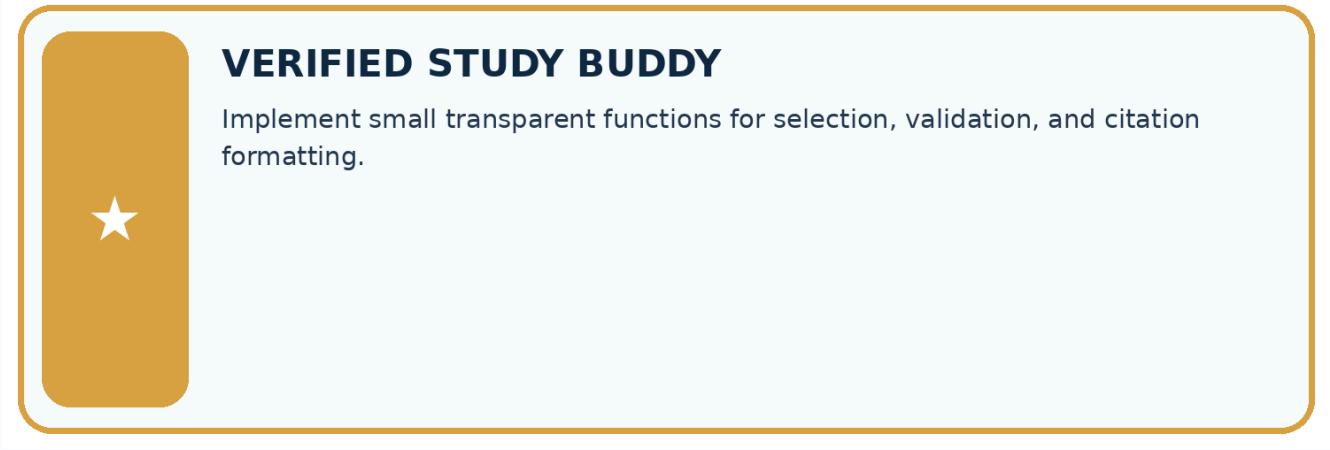
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

MAIN LESSON · SECTION 16

Finish: explain order, indentation and the evidence of a run

> A correct prediction and a successful execution are distinct records.

Explain without notes

- Close your notes. Explain this goal in your own words: Translate a short paper calculation into a Python script

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
variable	A named reference to a value that can be used or changed in a program.	
list	An ordered collection of items.	
function	A named unit of code that performs a task, optionally receiving arguments and returning a value.	
condition	A statement evaluated as true or false to help select an action.	

Terms for the connected explanation

Word	Meaning
Algorithm	An ordered procedure for solving a stated problem
Program	Instructions written in a language a computer can execute

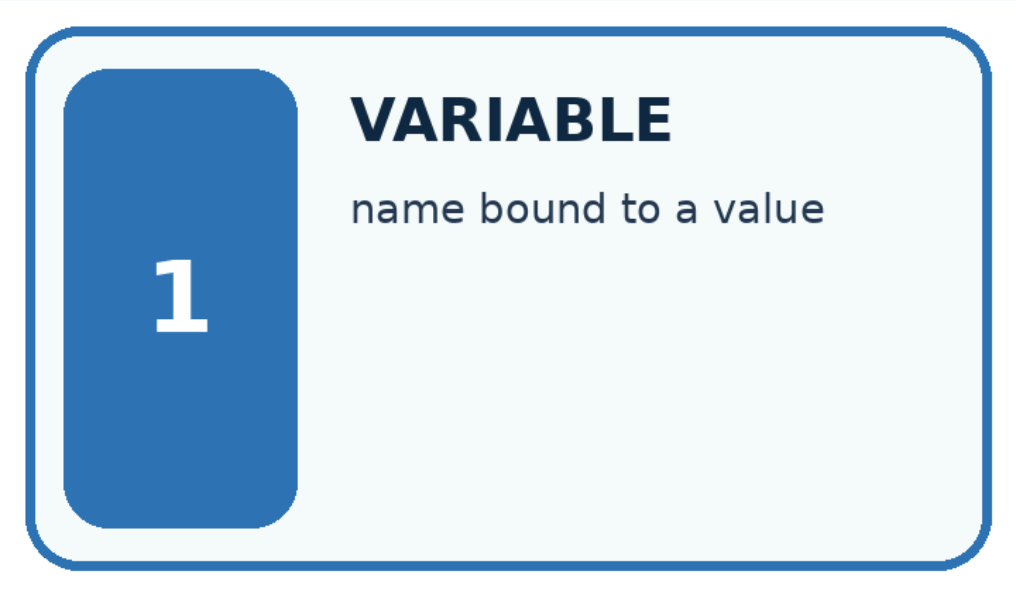
Word	Meaning
Interpreter	Software that executes Python instructions
Assignment	Binding a name to the value of an expression
Output	Information a program displays or returns

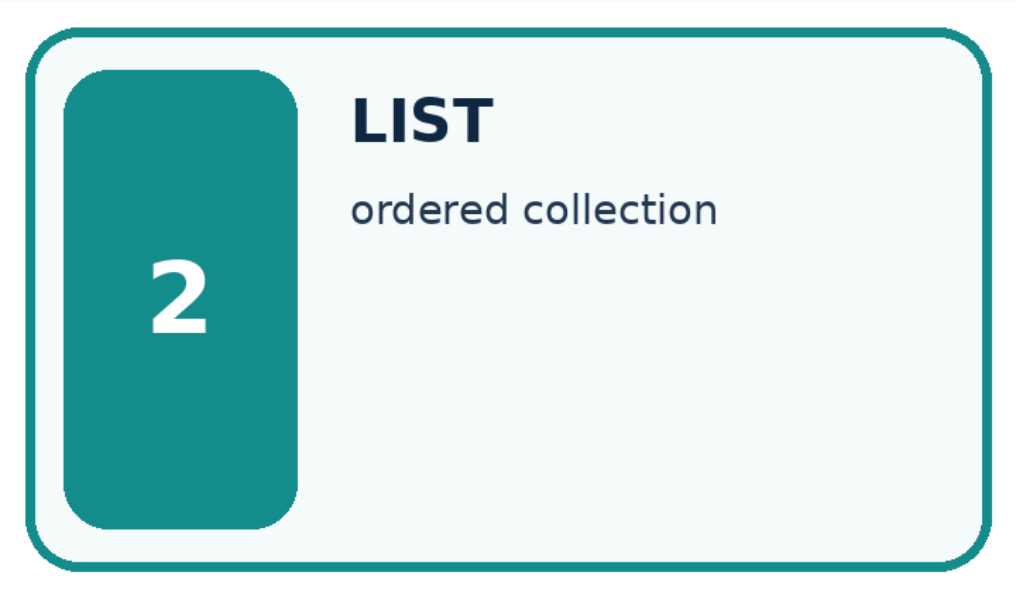
▼ Earlier wording, alternatives and picture transcripts

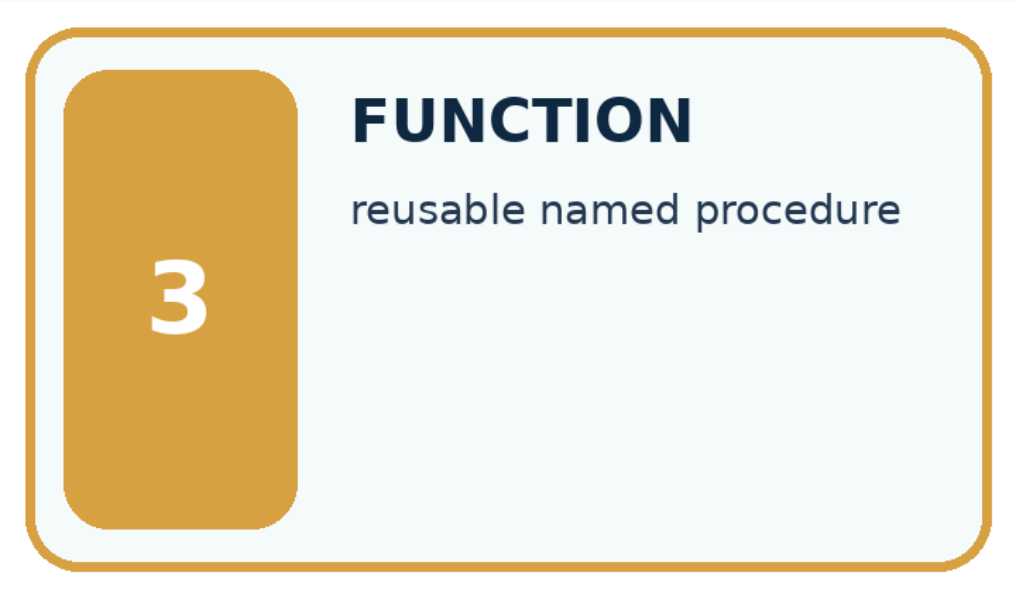
VOCABULARY LAB

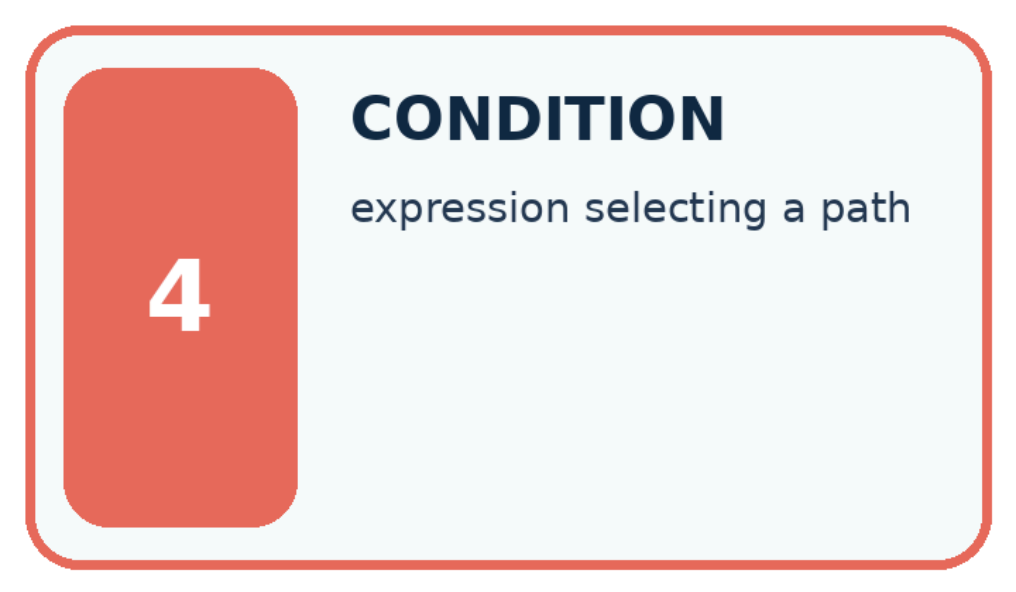
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
LIST
ordered collection


3
FUNCTION
reusable named procedure



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

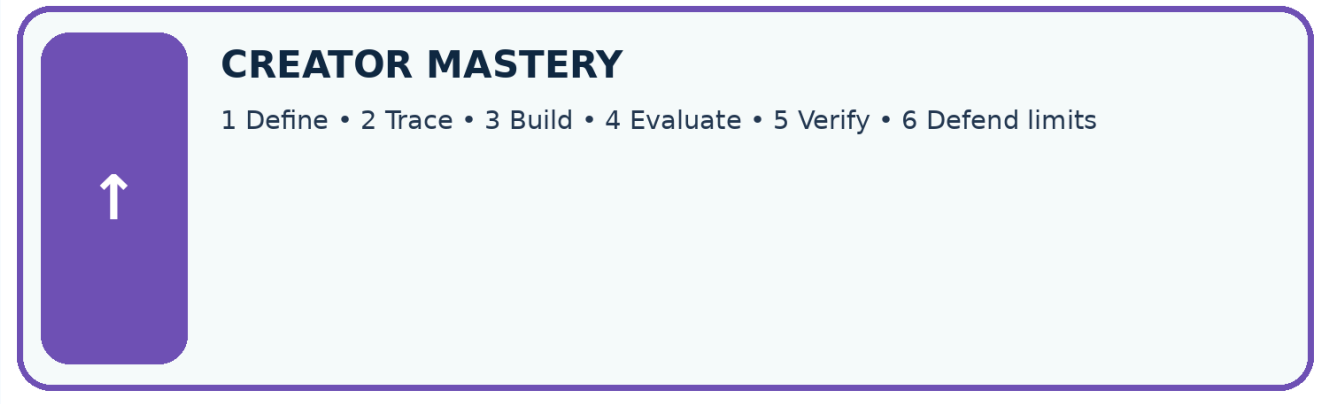
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Trace the new sum/reset procedure independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

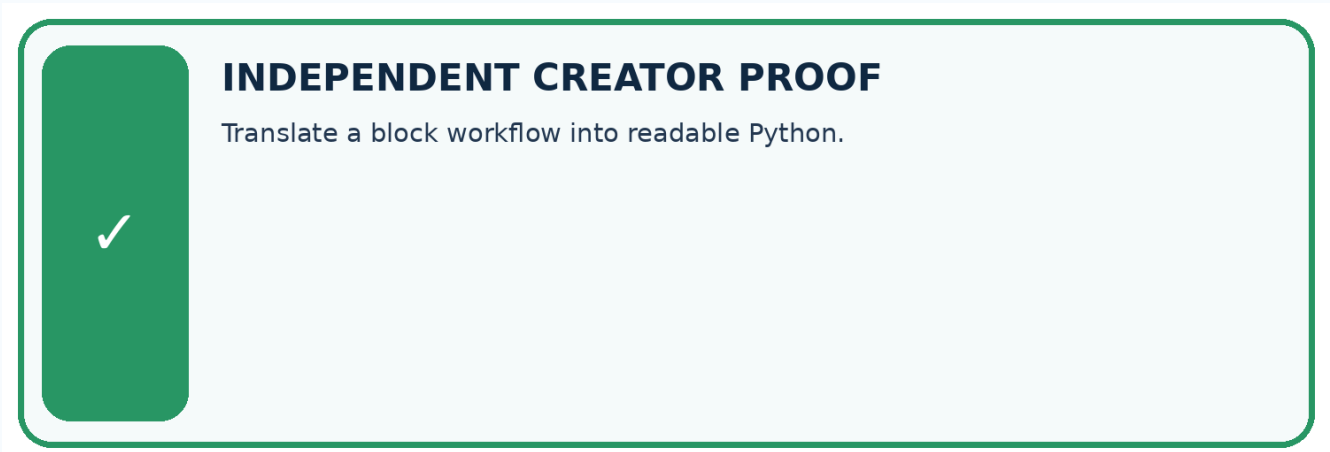
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Trace the new sum/reset procedure independently is the selected independent case.



EXPLANATION CHECK Require ordered intermediate state, initialization scope and separation of paper trace from executed output.

Write or draw your reasoning and evidence.

▼ **Extra practice: Trace the accumulator inside the loop**

REFERENCE · SECTION 13

Extra practice: Trace the accumulator inside the loop

> Try the question before opening its worked explanation.

Read the supplied case and question

A block **program** sets total=0 and adds each value in [2,4,6].

Translate and trace.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Python uses total=0, then an indented total += value inside for value in [2,4,6]. Totals are 2,6,12.

Evidence or stage	Value or result
Start total	0
Add 2	2
Add 4	6
Add 6	12

▼ Earlier wording, alternatives and picture transcripts

Week 23 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Trace the accumulator inside the loop

A block **program** sets total=0 and adds each value in [2,4,6].

Translate and trace.

Python uses total=0, then an indented total += value inside for value in [2,4,6]. Totals are 2,6,12.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Translate a short paper calculation into a Python script	Explain the deciding step, show a trace or test, and state one limit.
Predict and explain the order of executed statements	Explain the deciding step, show a trace or test, and state one limit.
Keep expected and observed output separate when checking a program	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Translate a “count qualifying books” paper procedure into a Python-style trace.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Translate a “count qualifying books” paper procedure into a Python-style trace.
- Which labelled step matches this detail? The final result is 2 qualifying books, not the sum of their page counts.
- 1. Trace decisions
- 2. Start counter
- 3. Report count
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Report count The final result is 2 qualifying books, not the sum of their page counts. Read the picture in order.

  The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: Replace the last page count 90 with 100. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The final count becomes 1.
- Hint if needed: Use the changed **condition** and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: The final count becomes 1. The comparison is strictly less than 100, so the boundary value does not qualify.

My selected conclusion, deciding fact and explanation

▼ Check the changed case

Repair a missing building block

Repair and save your work

Review Requests Responses and JSON (ai-3/22). Goal: Parse JSON into Python records.
Short practice: Would converting the text silently prove the source count is correct?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Conversation State and Memory (ai-3/20). Goal: Explain the **mechanism** and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Model Memory vs Retrieval (ai-3/17). Goal: Explain the **mechanism** and terminology of retrieval and indexing. Short practice: Why can the assistant not certify 15: 00 when the notice is missing?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Predict, run and debug

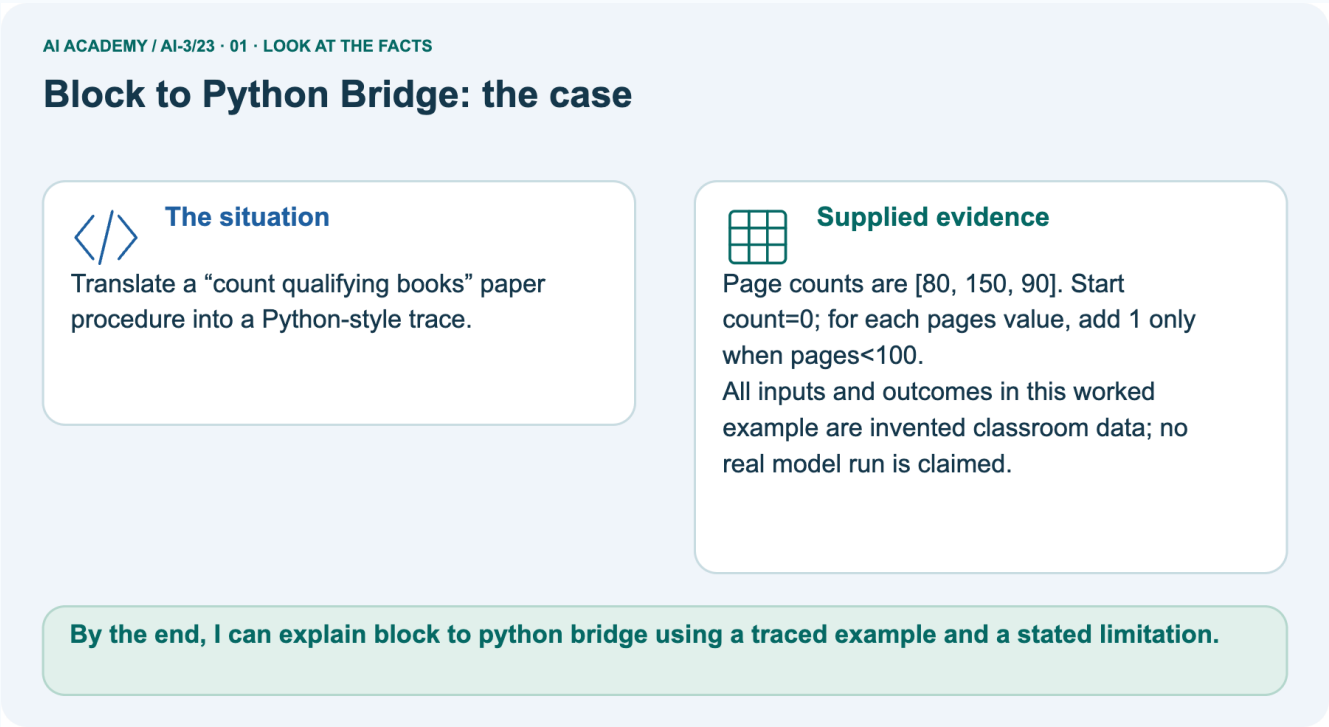
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Block to Python Bridge: the case



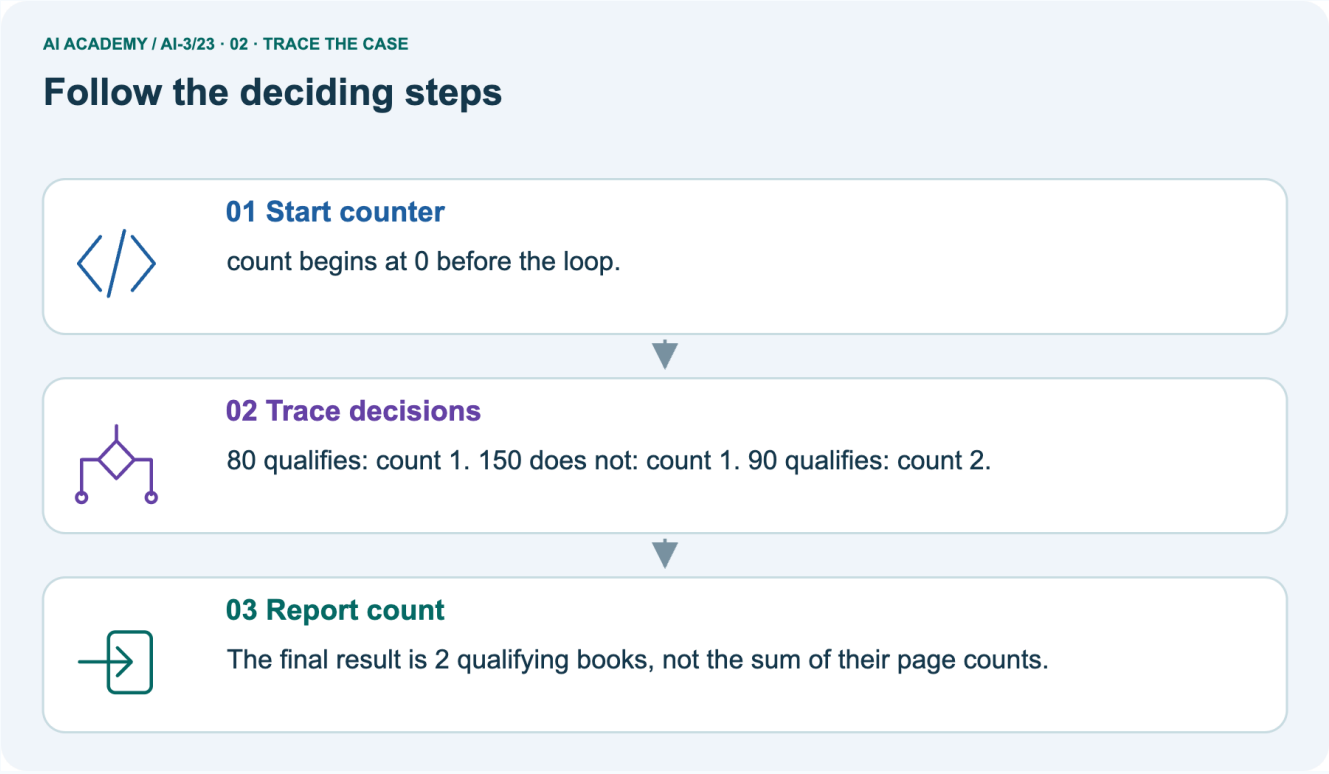
Follow the labelled picture with the matching explanation. Open the vector diagram

Translate a “count qualifying books” paper procedure into a Python-style trace. Supplied facts: Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

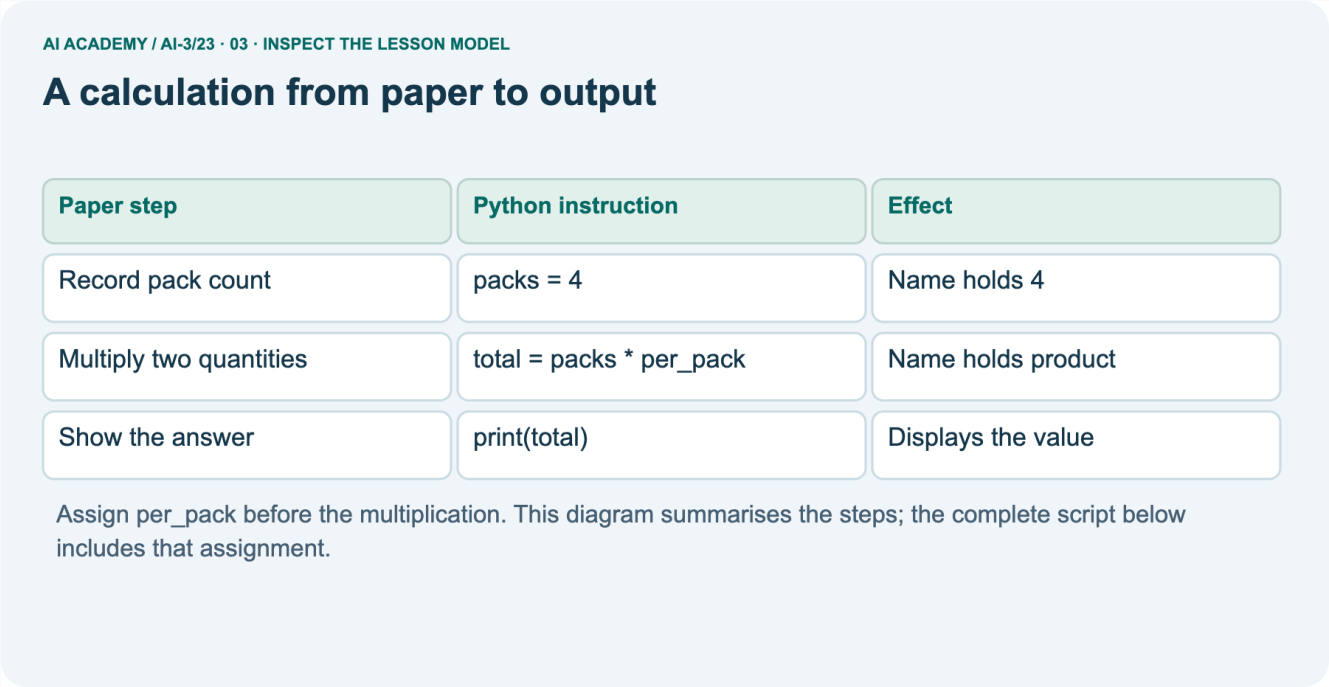


Follow the labelled picture with the matching explanation. Open the vector diagram

Start counter: count begins at 0 before the loop. → Trace decisions: 80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2. → Report count: The final result is 2 qualifying books, not the sum of their page counts.

Point to each numbered stage and explain which recorded **input** it uses.

03. A calculation from paper to output

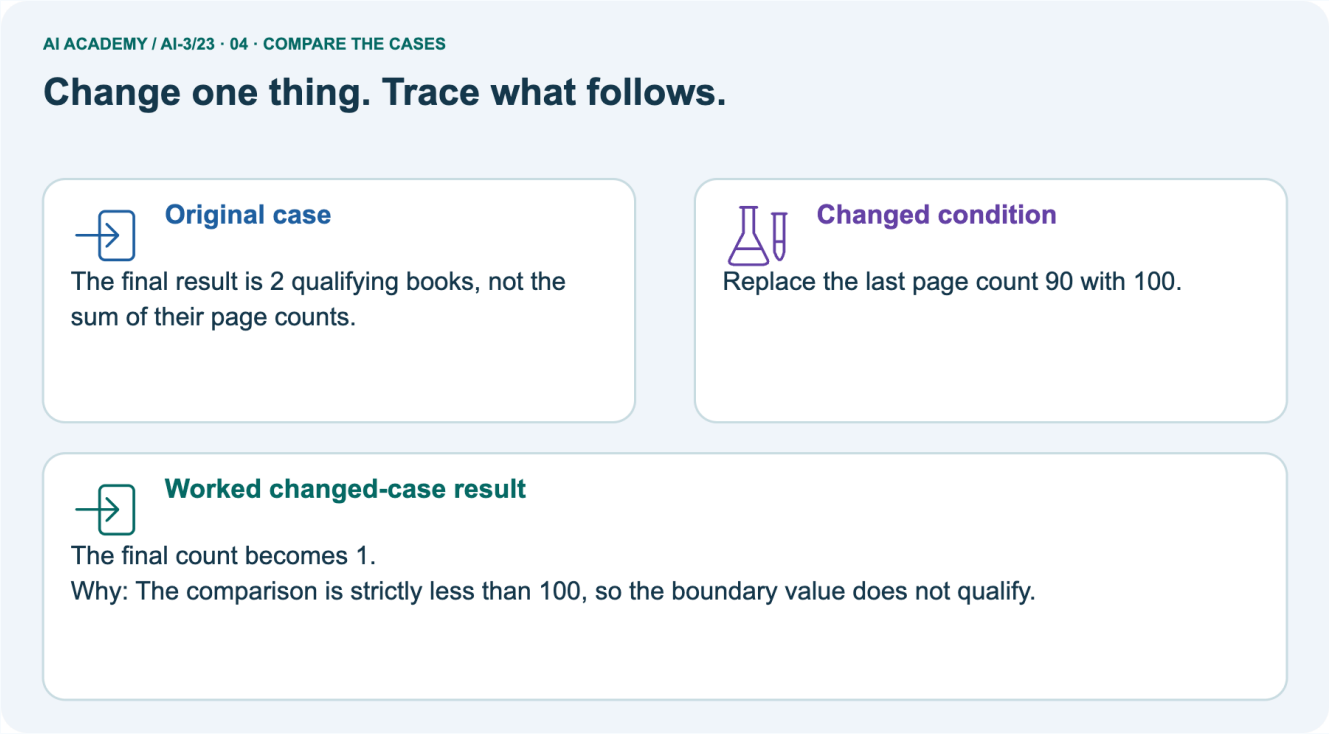


Follow the labelled picture with the matching explanation. Open the vector diagram

A calculation from paper to **output**. Record pack count | packs = 4 | Name holds 4. Multiply two quantities | total = packs * per_pack | Name holds product. Show the answer | print(total) | Displays the value

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



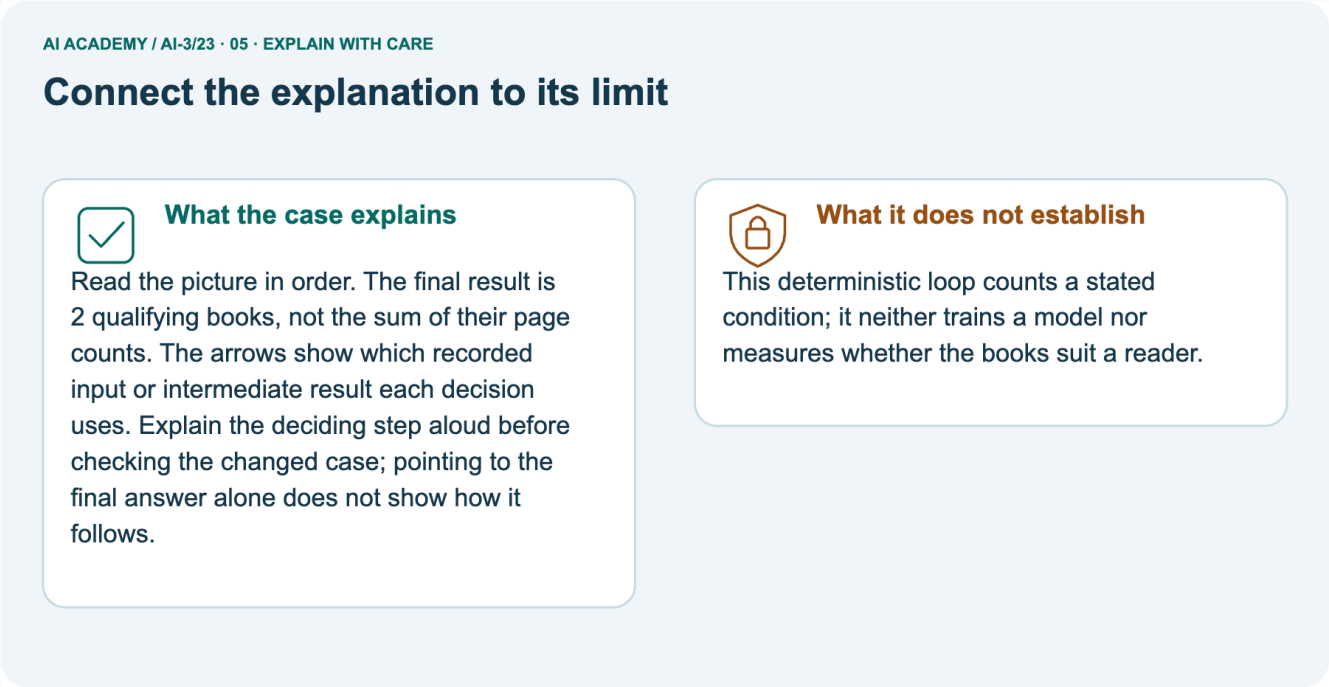
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The final result is 2 qualifying books, not the sum of their page counts. Changed **condition**: Replace the last page count 90 with 100. Expected result: The final count becomes 1.

Reason: The comparison is strictly less than 100, so the boundary value does not qualify.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

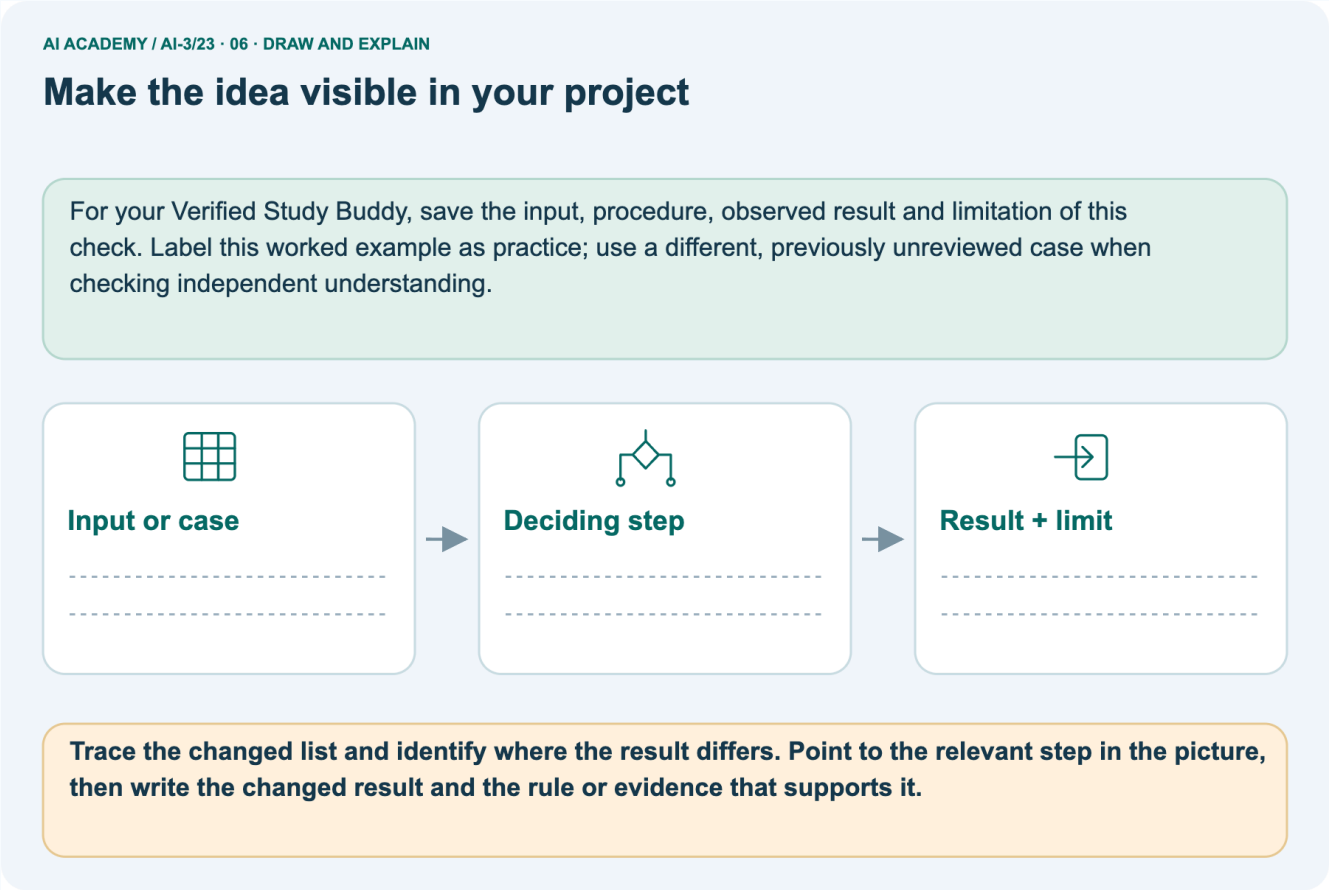
Read the picture in order. The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: This deterministic loop counts a stated **condition**; it neither trains a model nor measures whether the books suit a reader.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Trace the changed **list** and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Block to Python Bridge: the case

02. Follow the deciding steps

03. A calculation from paper to output

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.